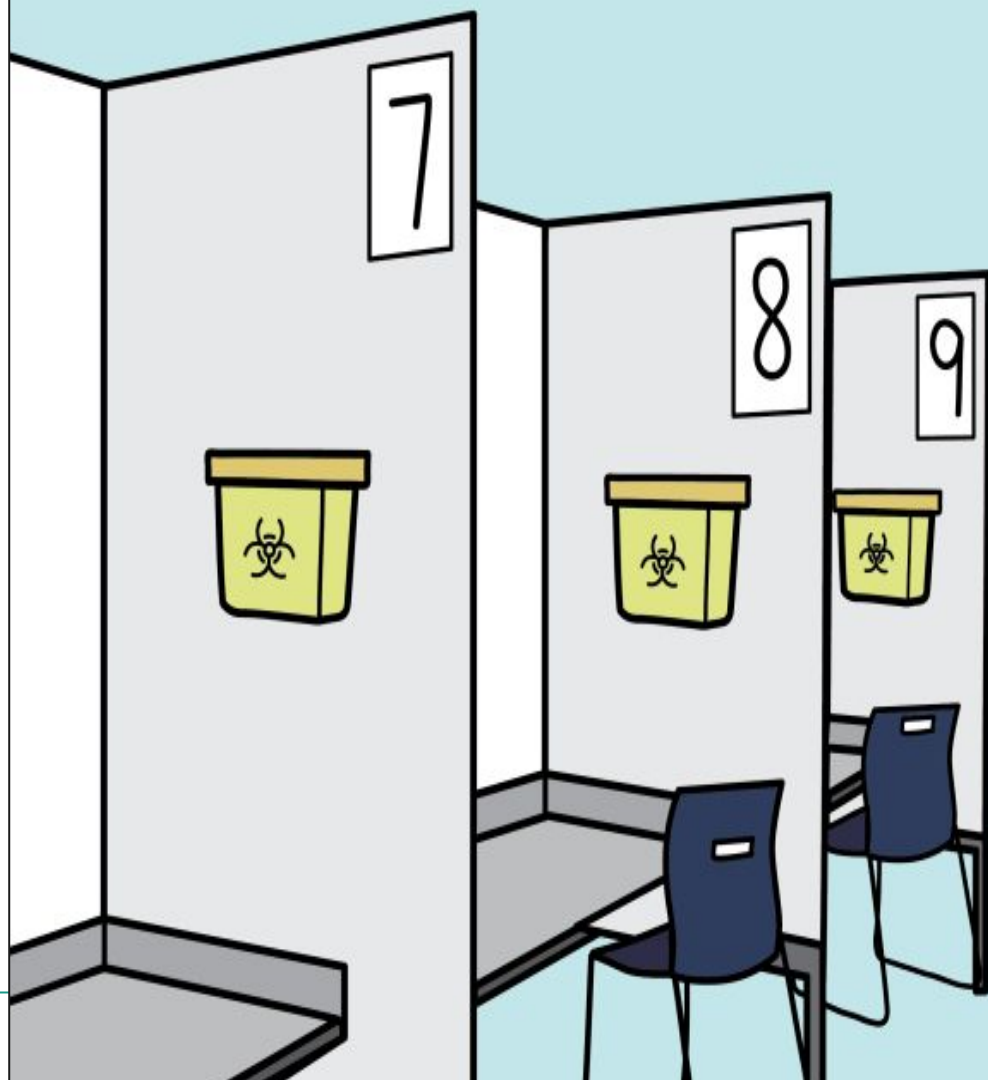


Mapping Service Gaps: A Spatial Analysis for Social Service Planning Amid Closures of Supervised Drug Consumption Sites in Toronto

A GIS-Based Analysis to Identify Areas Impacted
by Supervised Consumption Site Closures and
Recommend Outreach and Mitigation Strategies to
Reduce Overdose Risk

By Pravar Joshi



Executive Summary

Context:

In April 2025, several Supervised Consumption Sites (SCS) in Toronto were closed due to provincial policy changes. These closures risk leaving high-density, high-need communities with reduced access to life-saving harm reduction services.

Purpose of Analysis:

This project uses spatial and network analysis techniques, such as the ones below, to advise Social Services organizations:

- Quantify “crisis risk” due to SCS site closures by applying a Multi Criteria Decision Analysis approach
- Identify neighbourhoods and micro-areas most affected by using crisis risk score and other variables related to overdoses
- Explore how mobile health outreach can adapt to account for service areas lost due to site closures

Key Findings:

1. Service loss is not evenly distributed; the most severely impacted areas are also the most densely populated and overdose-prone
2. Traditional neighbourhood-level aggregation can hide disparities because healthcare service loss is often greatest in small pockets within broader communities
3. Mobile health outreach has the potential to restore access efficiently, particularly if route and stop locations are optimized based on insights from thematic analysis on which areas need the most help

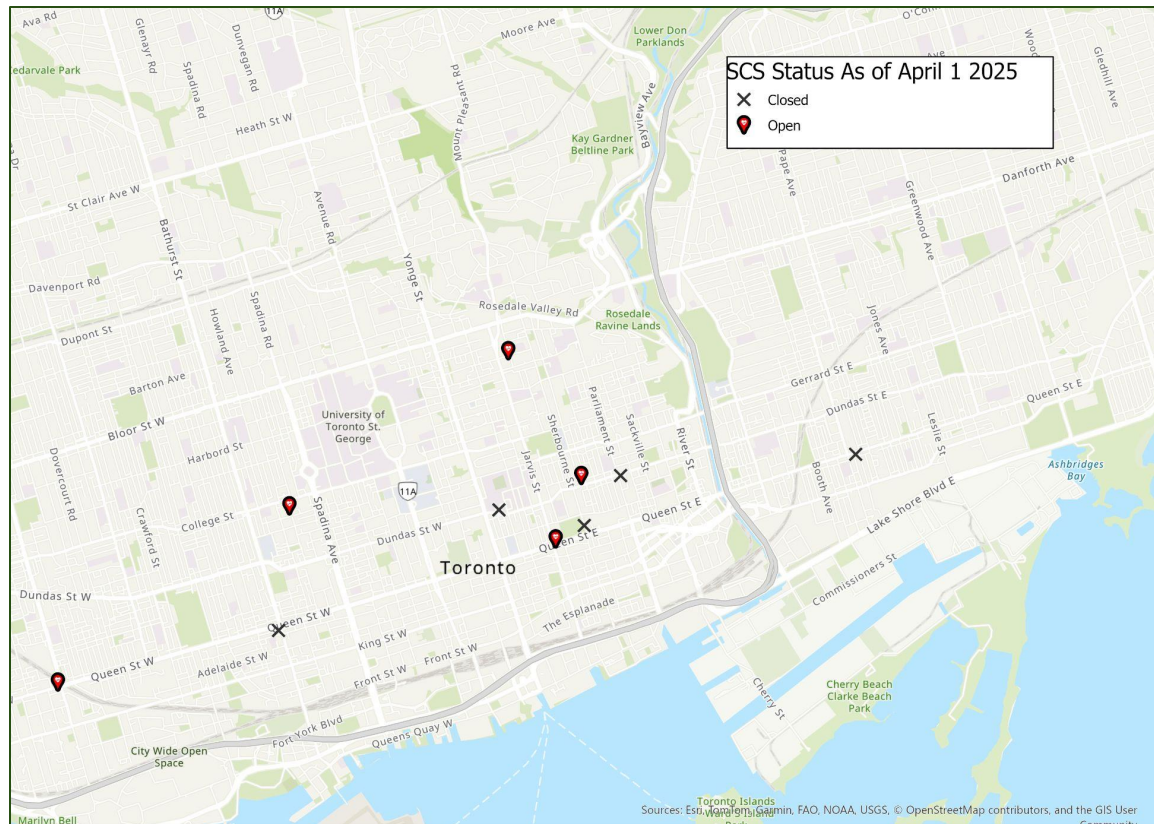
Introduction, Methodology, and Analysis Limitations

New Provincial Policy Leads to Closure of 5 Supervised Consumption Sites in Toronto

In 2024, Ontario implemented a policy requiring the closure of any **Supervised Consumption Site (SCS)** located within **200m of a school**

This policy led to the closure of **five SCS locations in Toronto**, disproportionately affecting the city due to its high-density

These sites provided **safe environments for drug use** to prevent overdose; closures now create **critical gaps in harm reduction services**



Map of active and closed SCS sites in Toronto following the 200m school proximity policy

Scoring Neighbourhood Risk Using Multi-Criteria Analysis To Account for Complexity and Ambiguity In This Sensitive Public Health Topic

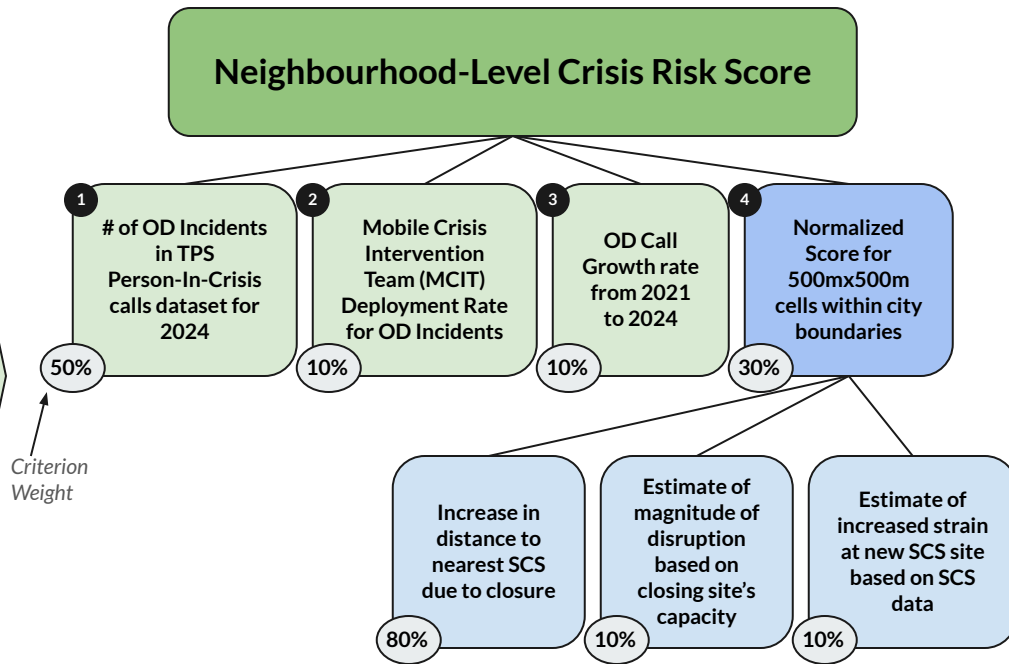
To address the risk of site closures, so that insights can inform decision making for social services organizations, it is critical to apply techniques from the **Multi-Criteria Decision Analysis framework**

A normalized **Crisis Risk Score** was calculated based on 2021 census-defined neighbourhood boundaries to account for:

1. Recent volume of Overdose-related calls
2. Involvement-level of mental health professionals to support OD incidents
3. Growth in calls over a period of time
4. Loss of site access within more granular areas of neighbourhood (another custom risk score)

Each variable was **normalized and weighted** based on relative importance to overdose risk (*refer to exact % values on the right*)

The final score was **mapped using a choropleth** to highlight neighbourhoods at highest risk



Modeling Travel Impacts Using Network and Routing Analysis

Network Analyst tools were used to assess how closures affect clients and mobile health services.

Network Dataset from Toronto's road and pedestrian layers was created with distance/time impedance for 2 analyses:

1. Simulated Client Travel Paths:

- Ran **Closest Facility Analysis** for incident points (representing 3 SCS clients) to nearest sites
- Created an animation with **Temporal Analysis** showing how walking routes changes for the 3 SCS clients pre-closure (March 31 2025) and post-closure (April 1 2025)

2. Mobile Health Clinic Routing

- Ran **Closest Facility Analysis** and **Routing Analysis**, combined with insights from **Crisis Score Choropleth** to model routes for Parkdale Queen West CHC's mobile health clinic based on realistic travel constraints (e.g., existing schedule)



This approach allows service providers to adapt mobile health routes based on where access loss is greatest

Data Limitations and Methodology Constraints in the Analytic Techniques

Data Limitations

Use of Proxy Data:

Overdose-related 'Person in Crisis' calls from Toronto Police are used as a proxy for where people who use drugs live. This significantly underrepresents the actual population and thus is meant to use for directional insights only.

Lack of Real-Time Data:

The analysis uses static snapshots of overdose call data and site status as of specific dates. This limits the ability to detect trends over time, such as seasonal spikes in overdoses or changing patterns of SCS usage following a closure (although it was reported that some open sites are seeing increased capacity as of April 1 2025)

Methodology Constraints

Thematic Mapping Aggregation Issues:

While creating a smaller area of 500m x 500m cells helps mitigate the Modifiable Areal Unit Problem (MAUP), the neighborhood-level risk score still aggregates diverse realities, potentially masking hyperlocal trends affecting vulnerable populations..

Network Assumptions and Impedances:

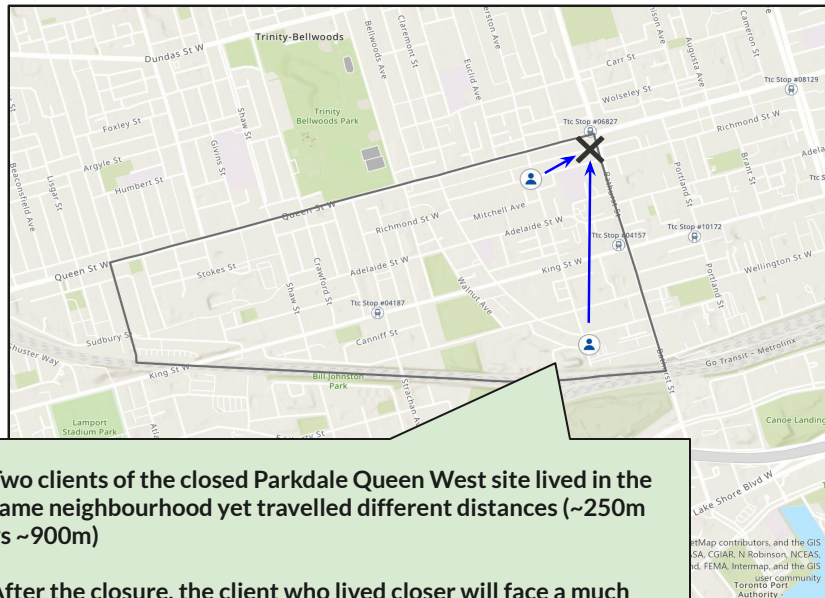
Network Dataset assumes optimal travel conditions and doesn't account for weather, construction, or stigma-related avoidance of certain roads that can only be obtained through lived experience

Thematic Mapping Analysis

Techniques Applied

- *Choropleth Mapping*
- *Dasymetric Mapping*
- *Graduated Symbolology*
- *Spatial Joins*
- *Custom Fields*
- *Clipping Analysis*

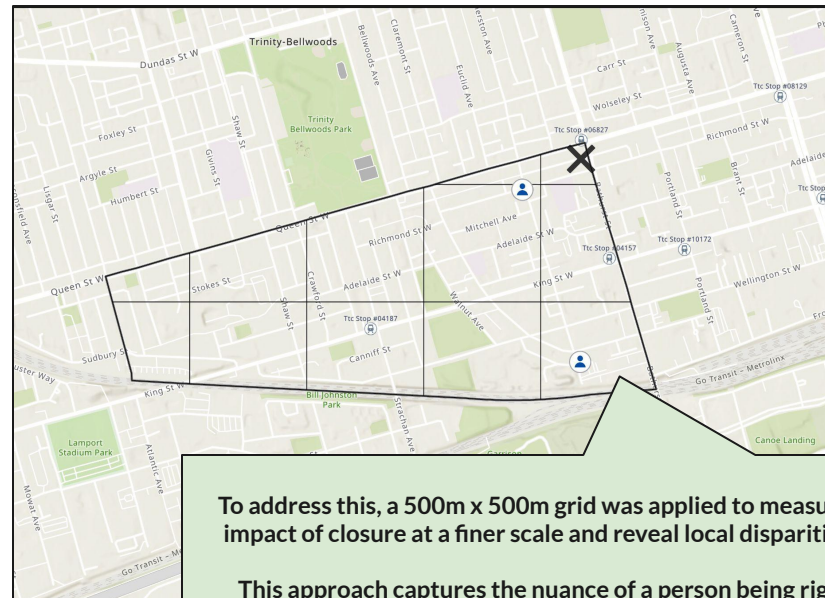
Same Neighbourhood, Different Realities: Why Aggregated Data Misses Gaps & How Fine-Scale Analysis Helps



Two clients of the closed Parkdale Queen West site lived in the same neighbourhood yet travelled different distances (~250m vs ~900m)

After the closure, the client who lived closer will face a much longer trip, creating a new barrier to access

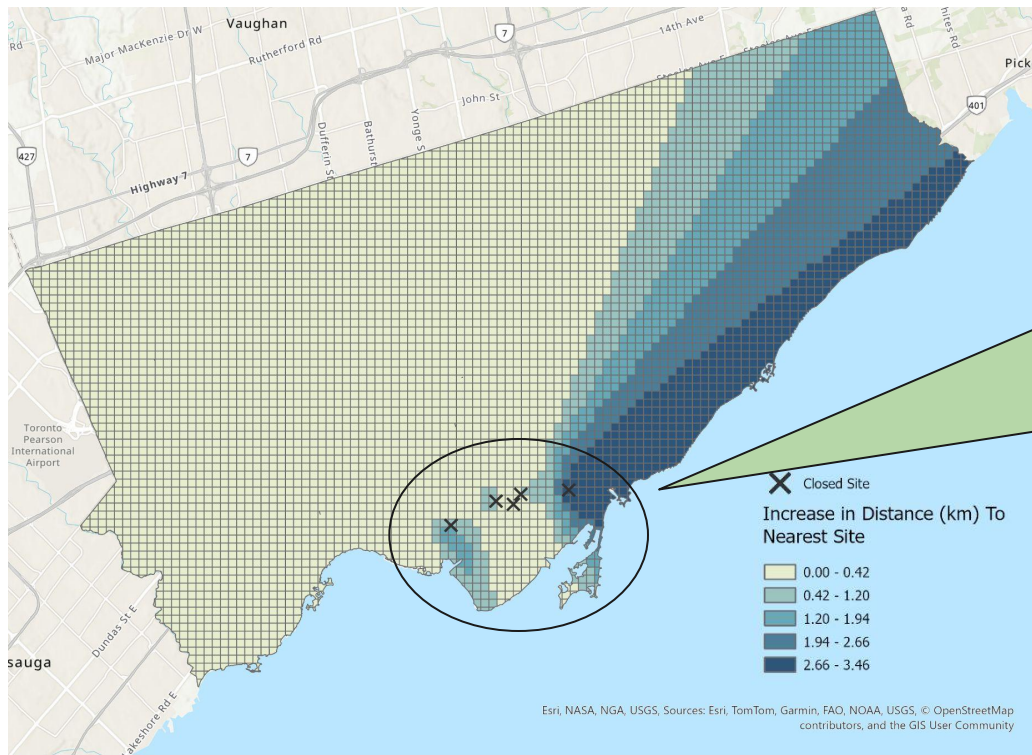
These unequal impacts are masked by neighbourhood level analysis and can increase health risks like overdose



To address this, a 500m x 500m grid was applied to measure impact of closure at a finer scale and reveal local disparities

This approach captures the nuance of a person being right next to a former site being more impacted, since they have to travel further than their usual route.

Localized Disruption: Mapping Travel Distance Increases Across Toronto



Distance increases are most severe in downtown micro-areas where SCSs were previously concentrated. Though small in land area, these zones have high population and overdose density.

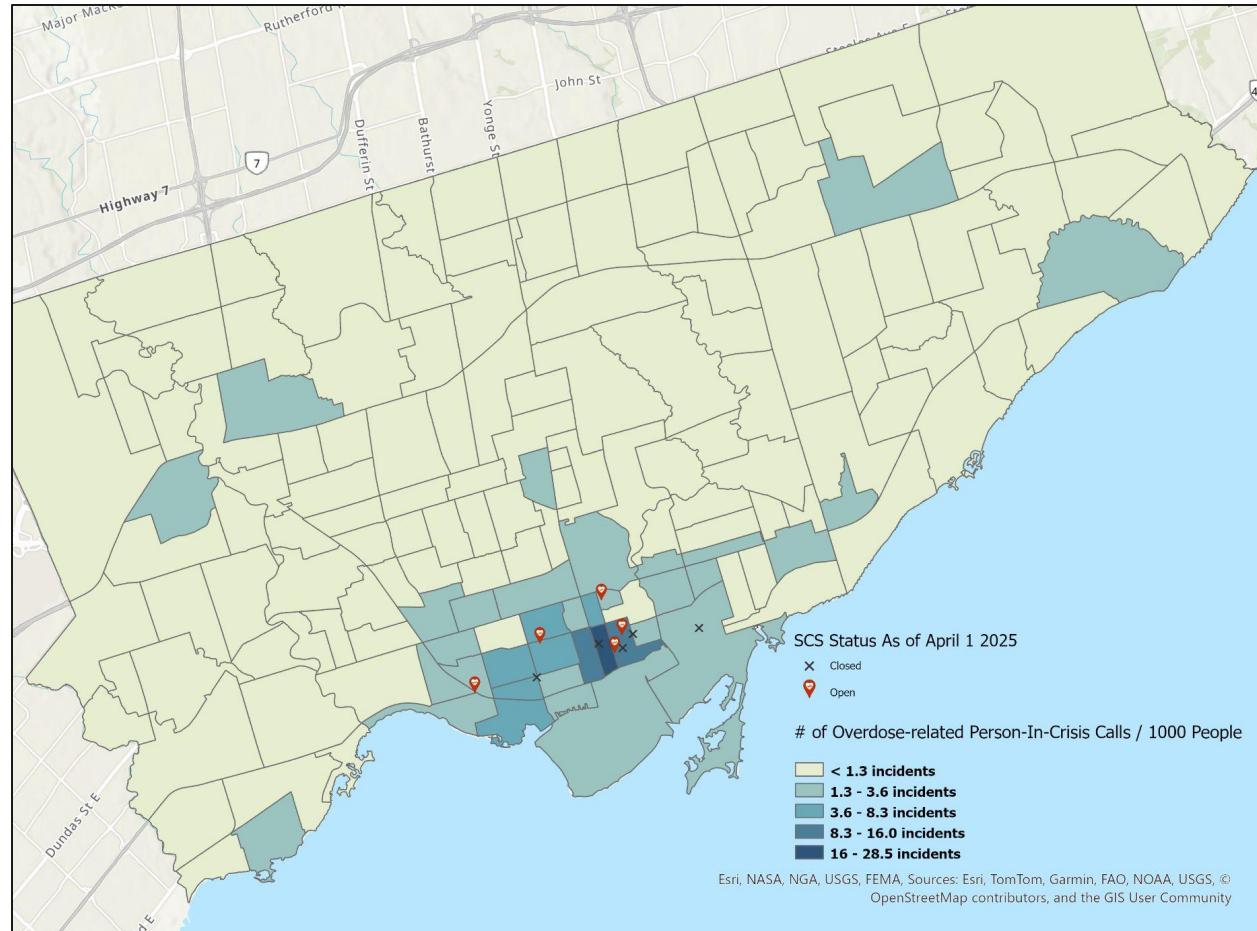
Due to the loss of South Riverdale Community Health Centre in the East End of the City, residents living in the area had a uniform increase in distance to the nearest site although their prior distance was already high (the further away from the site you go)

Sites Closed Where Overdose Incidents Increase

The **dark shaded neighbourhoods** (8.3–28.5 calls/1,000) are all located downtown, including Moss Park, Kensington-Chinatown, and Waterfront communities.

These **areas coincide exactly with where multiple SCSs have recently closed**, leaving the highest-need communities most exposed.

The **lack of nearby sites** for the handful of high-volume neighbourhoods **away from downtown core** is also a cause for concern

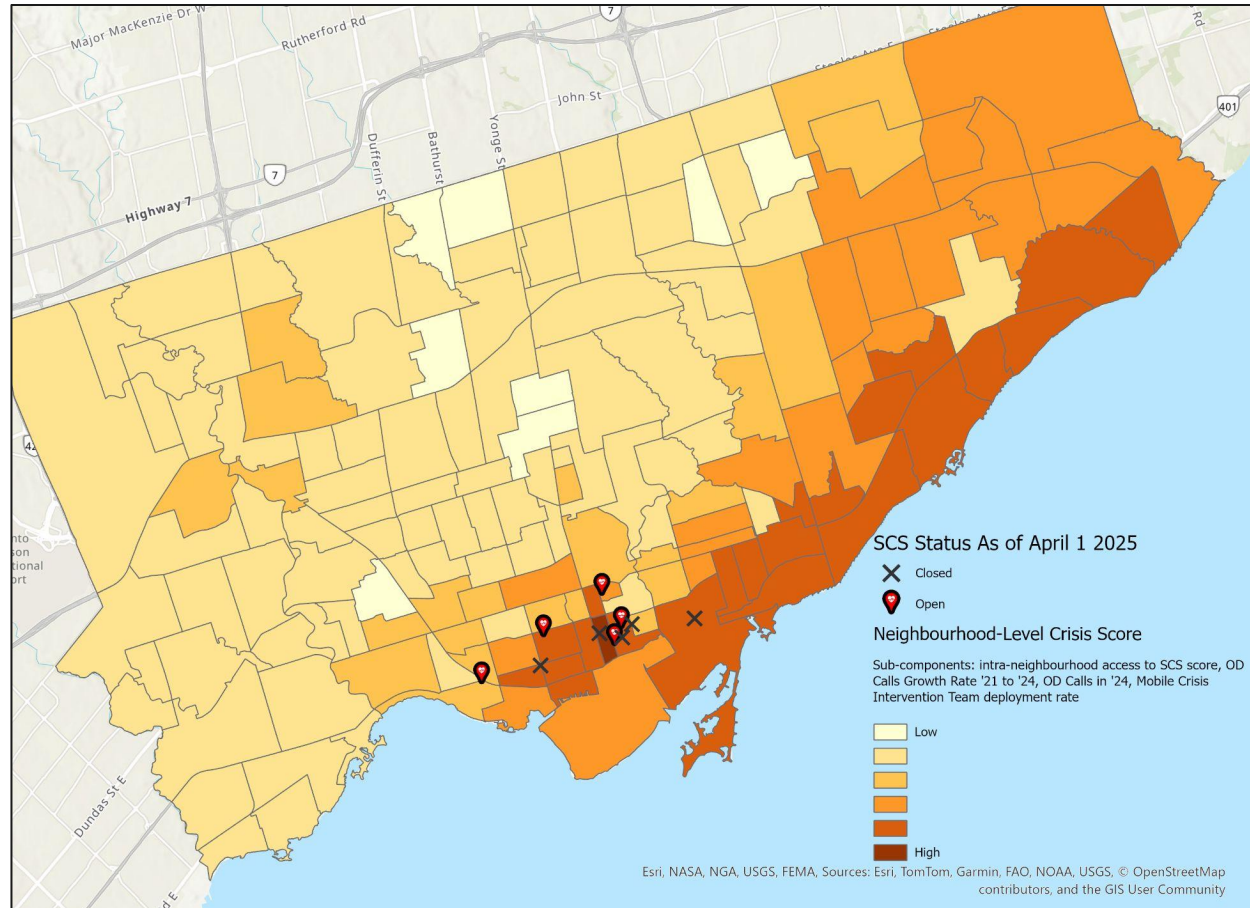


Neighbourhood Level Composite Crisis Score

Both maps confirm: **Downtown Toronto** (e.g., Moss Park, Waterfront, Trinity-Bellwoods) **faces the greatest** overdose-related **risks**.

While the downtown core remains a focal point, the **Crisis Score identifies additional vulnerable neighbourhoods** that may have been missed in a single-variable analysis:

- Parkdale-High Park
- Eastern Lakefront
- South Riverdale
- Dovercourt-Wallace



Network Analysis

Techniques Applied

- *Building Network Dataset*
- *Closest Facility Analysis*
- *Routing Analysis*
- *Temporal Analysis & Animation*
- *Integrating with Thematic Mapping*

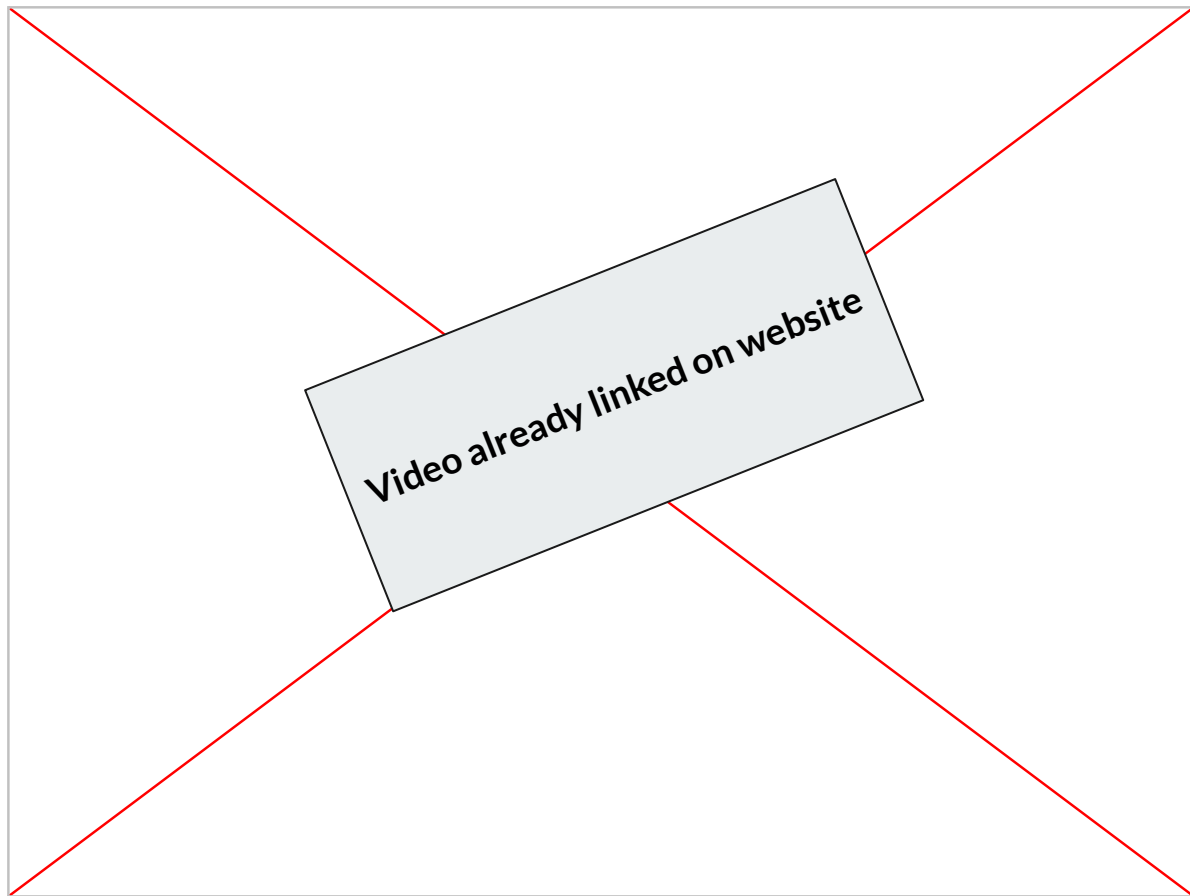
When Sites Close, Clients Travel Farther: A Temporal View of Access Loss

Animation shows a **temporal analysis** with 3 randomly generated **SCS clients' travel paths before and after April 1 closures**

Site closures force clients to walk farther, increasing time, effort, and wait times

Even small geographic shifts can significantly affect **vulnerable individuals** as they **do not have the same access to faster and more convenient transportation** options like cars, taxis, and public transportation.

The next part of this analysis investigates **how care can be delivered directly** to where individuals are located



Parkdale Queen West Mobile Health Clinic: Bringing Care to Where It's Needed Most

In 2021, **Health for Good** was launched through a partnership with TELUS Health, Parkdale Queen West CHC, and UHN's Social Medicine Program **to deliver healthcare services out of a specially-equipped vehicle.**

Delivers primary care, harm reduction, and COVID-19 response services to underserved populations in Toronto's midwest neighbourhoods

Aims to reach individuals facing barriers to traditional care — including those experiencing homelessness or living in poverty



Eligibility

Open to everyone

Languages of Service

Services are provided in English.
Interpretation is available if needed.

Source: <https://pqwchc.org/programs-services/health-care/telus-mobile-health-clinic/>

Visualizing The Areas Served By The Mobile Health Clinic To See What To Improve

The original **Parkdale CHC catchment area (green)** reflects the intended geographic scope of in-person services prior to the expansion of this mobile health initiative.

The **expanded catchment area (blue)** demonstrates how mobile health **routing** extends service access well beyond the traditional boundary, which bodes well for hopefully extending service areas to support harm reduction

Week 1 mobile clinic stop locations are mapped to illustrate the spatial distribution and frequency of outreach efforts during the first week of the operating cycle (pg 21 has Week 2 stops)



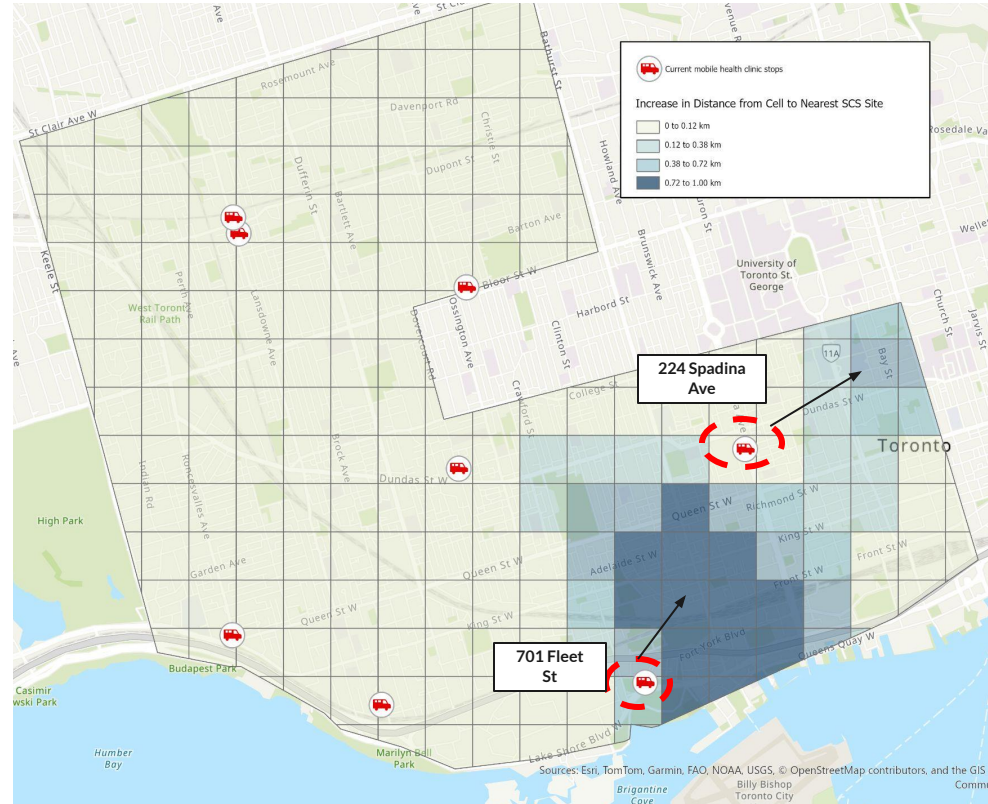
Targeted ‘Micro’-level Adjustments to Improve Mobile Service Reach

The choropleth visualizes the increase in walking distance to the nearest Supervised Consumption Site (SCS) after April 1 closures, clipped to the mobile clinic’s current catchment area.

All existing clinic stops are mapped with 2 recommendations:

- Suggested **additional stop northeast of 224 Spadina Ave** location to serve area of SCS closure.
- Another stop suggested **slightly northeast of 701 Fleet** to serve the most impacted (by distance) areas in Toronto
-

If additional funding/capacity exists, **these minor route modifications should be prioritized** to help serve vulnerable communities impacted by SCS closures

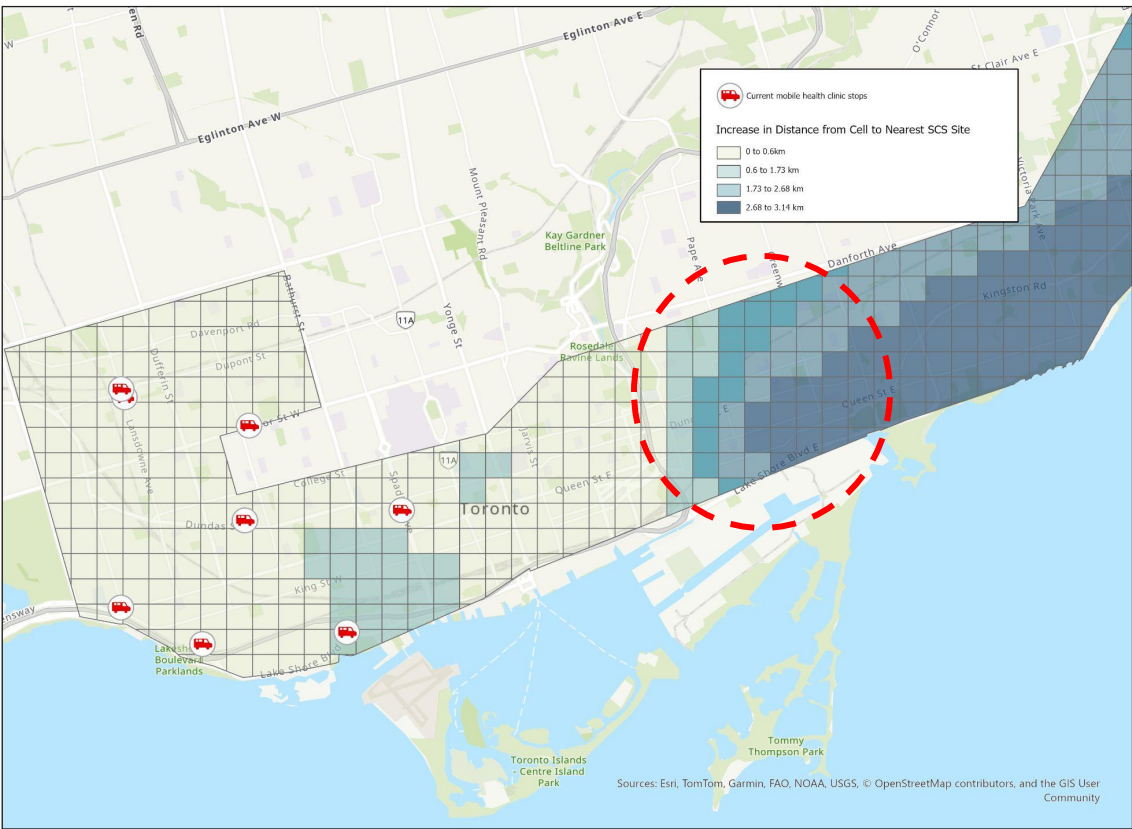


Expanding Reach: Addressing The Service Gaps Created in Toronto's East End Following Site Closure

The **catchment overlay** was removed to show **disparities in the east end** , where service loss is high due to closure of site in South Riverdale

This area currently falls outside the mobile clinic's typical operating zone despite high vulnerability

Should resources allow for expansion of the mobile clinic services to the weekend, then the **South Riverdale neighbourhood should be prioritized**



Conclusion

Recent closures of these SCS sites have made it harder for people who use drugs to reach life-saving services:

- Parkdale Queen West Community Health Centre (Queen West Site)
- Moss Park Consumption & Treatment Service
- The Works – Toronto
- South Riverdale Community Health Centre
- Regent Park Community Health Centre

The impact is not equal with some areas hit much harder than others, especially in the east end and downtown. Mapping and routing analysis shows that existing mobile health clinics could provide relief to overburdened sites that remain open with some scheduling tweaks.

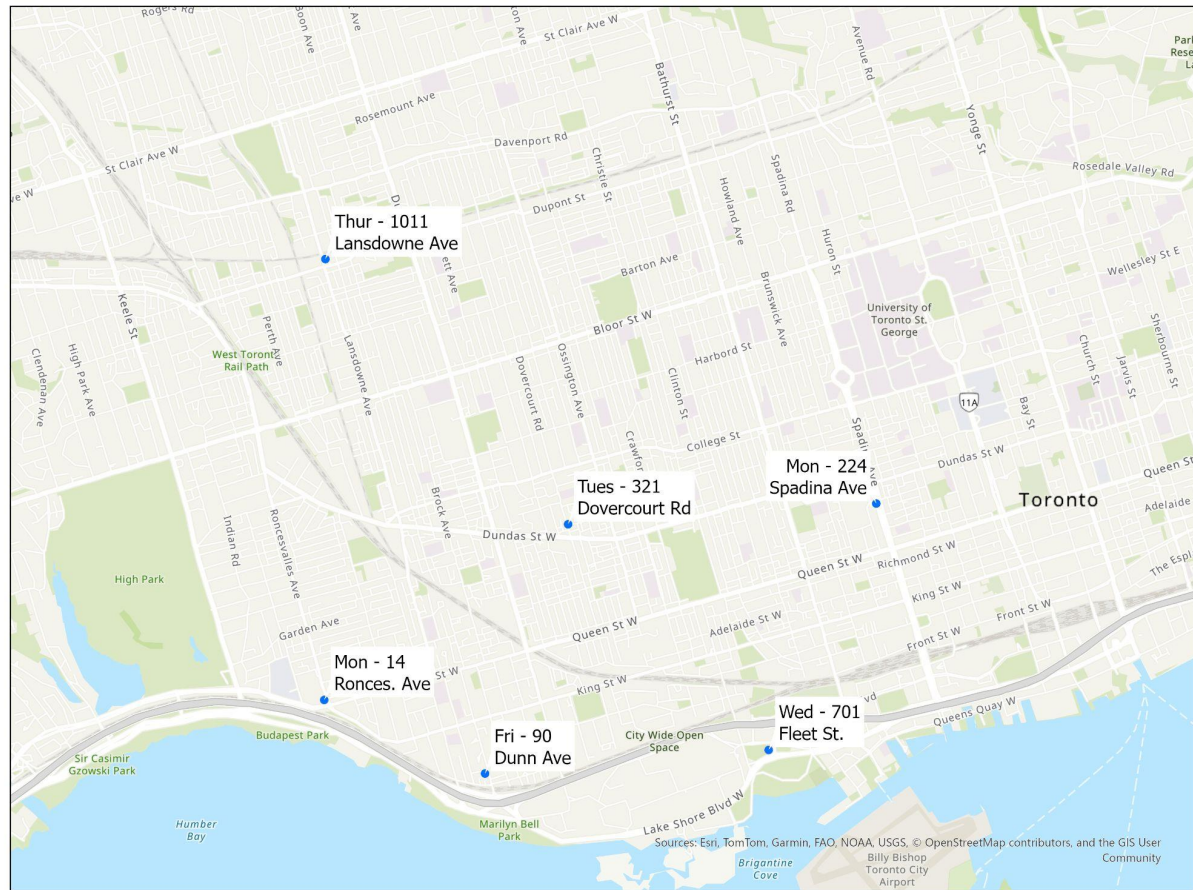


Summary of Recommendations

1. Rework outreach routes to better cover areas that lost access, especially in the east end near South Riverdale and downtown east of Spadina (essentially spots right next to closed sites)
2. Add a new mobile clinic (Parkdale Queen West CHC-led) stop near South Riverdale CHC if possible
3. Adjust current stops (e.g. 224 Spadina and 701 Fleet) slightly to better reach nearby high-need areas
4. Include distance to services as a key factor in any future overdose risk or service access model
5. Disseminate pamphlets to clients of closed sites that provide walking directions to nearby sites and/or distribute TTC Presto Passes to expedite travel without adding cost burden

Appendix

Appendix - Mobile Health Clinic's Week 2 Stops



Datasets Used

1. Toronto Police Service (TPS) Person-in-Crisis Calls Dataset (2024)
2. Supervised Consumption Site (SCS) Locations and Dates from:
<https://www.cbc.ca/news/canada/toronto/closed-supervised-consumption-site-impact-1.7551337>
3. SCS Healthcare data from: <https://health-infobase.canada.ca/supervised-consumption-sites/>
4. Toronto Census 2021 Data
5. Toronto Road Network and Pedestrian Layer
6. Parkdale Queen West Mobile Health Clinic Route and Schedule Data from:
<https://pqwchc.org/programs-services/health-care/telus-mobile-health-clinic/>